



SECD2613 SYSTEM ANALYSIS AND DESIGN

Assignment 1: Project Planning

Clinic Appointment and Patient Queue Management System

Semester	Semester II 2025/2026
Type	Group Assignment (5%)
Duration	1 week
Submission	Submit softcopy in PDF via e-learning

Feasibility Study

Scenario

MediCare Plus Clinic currently manages appointments using phone calls, WhatsApp messages, paper appointment books, and Excel files. As the number of patients increases, the clinic faces duplicate booking slots, difficulty tracking patient queue status, long waiting times, delayed reporting, and inefficient access to appointment information. The clinic is considering a Clinic Appointment and Patient Queue Management System, but before making the investment, management requires a proper feasibility study and project planning analysis.

Student Tasks

Item	Task	Required Output
1	Problem Definition	Write under these exact subheadings: (a) Current process, (b) Current problems, (c) Formal problem statement. Include at least 4 distinct current problems.
2	Objectives, Requirements, and Constraints	Write under these exact subheadings: (a) Objectives - minimum 4, (b) Functional requirements - minimum 6, (c) Non-functional requirements - minimum 4, (d) Constraints - minimum 4. Use numbered lists.
3	Feasibility Study	Write under these exact subheadings: (a) Technical feasibility, (b) Operational feasibility, (c) Economic feasibility, (d) Overall feasibility decision. Each feasibility subsection must end with a clear verdict: Feasible / Partially feasible / Not feasible.
4	Economic Feasibility: Cost-Benefit Analysis (CBA)	Complete the provided 3-year CBA table. Show all numerical answers in the table. Use the Working / Formula column for at least one sample formula per major section. State total PV costs, total PV benefits, gain/loss, profitability index, and justification.
5	Conclusion	Add a short final conclusion summarising whether the project should proceed and why, based on feasibility and project planning outputs.

Economic Feasibility (CBA)

Estimated Cost	
Hardware	RM 15,000.00
Software	RM 20,000.00
Maintenance	RM 5,000.00 / year
Support & Upgrade	RM 10,000.00 / year
Expected Benefits	
Interest investment	RM 1,000.00 / month
Consultation services	RM 20,000.00 / year
Assumptions	
Discount rate	8%
Sensitivity factor (cost)	1.1
Sensitivity factor (benefits)	0.9
Annual increase in production costs	5%
Annual increase in benefits	4%

Based on details above:

1. Calculate the cost-benefit estimation using the Present Value (PV) analysis (for 3 years) to assess the economic feasibility.
2. What is the PI value for this PV analysis? What is your recommendation based on the PI value and justify it.

Assessment Rubric

Criterion	Excellent (5)	Good (4)	Satisfactory (3)	Weak (2-1)	Unsatisfactory (0)
1. Problem Definition	All required subheadings present; 4 or more clear problems; strong formal problem statement linked to scenario.	Most subheadings present; 3-4 relevant problems; adequate problem statement.	Some required structure present; limited or repetitive problems; weak problem statement.	Missing structure; very few problems; unclear or irrelevant problem statement.	No Problem Definition
2. Objectives, Requirements, and Constraints	Meets or exceeds all minimum counts; items are relevant, distinct, and correctly classified.	Mostly meets counts with minor misclassification or overlap.	Below one or two minimum counts or several vague items.	Major omissions, wrong classification, or very limited coverage.	No Objectives, Requirements, and Constraints
3. Technical Feasibility	Addresses technology, infrastructure, skills/vendor issue, and ends with justified verdict.	Covers most required points with acceptable verdict.	Basic discussion with limited justification or unclear verdict.	Weak discussion or no clear verdict.	No Technical Feasibility
4. Operational Feasibility	Addresses workflow fit, staff acceptance, training/change issues, and ends with justified verdict.	Covers most required points with acceptable verdict.	Basic discussion with limited justification or unclear verdict.	Weak discussion or no clear verdict.	No Operational Feasibility
5. Economic Feasibility / CBA	All major values and final indicators are correct and internally consistent.	Minor arithmetic or carry-forward errors only.	Several errors but method is partly correct.	Major errors, incomplete table, or missing conclusion.	No Economic Feasibility / CBA